Action Principle

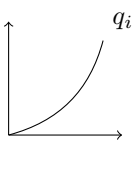
Recall Fermat's Principle: Light travels along the shortest path between 2 points.

Precisely: *length of a curve* $L = \int ds \sqrt{\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2}$ will be minimized for the trajectory of light.

- Every mechanical system obeys a minimization principle of this sort.

$\rightarrow$ Consider the set of all possible paths $q_i(t)$ that a system could take through configuration space.

Define the action if a path $q_i(t)$: $S[q_i(t)] = \int_{t_i}^{t_f} L(q_i, \dot{q}_i, t)$ $(q_i, \dot{q}_i \Rightarrow$ the path we choose)



(configuration space / space of all q_i's)

depends on $q_i(t)$ (function of a function)

The path through configuration space that a mechanical system takes "minimize" $S[q_i(t)]$ (minimize implies "local extremum")

This is called the least action principle or Hamilton's principle.

$S[q_i(t)]$ is a function of a function

"Functionals": use [ ] rather than ( ) to denote dependence.

Calculus of variations: We have used in single (or multi) variable calculus to minimizing a function of 1 (or n) variables.

We need to minimize a *functional* $\Rightarrow$ a function of an ∞ # of variables.

$S[q(t)] \xrightarrow{\text{infinit set of variables}}$